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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination - APRIL 2021

CH511S – EQUILIBRIUM THERMODYNAMICS OF GASEOUS SYSTEMS

Time : 3 hrs **Max Marks :75**

SECTION – A (15X1=15)

Answer **ALL** Questions

I. Choose the correct answer

1. Two moles of an ideal gas expand spontaneously into a vacuum. The work done is
   a) 2J b) Infinity c) Zero D) 3j

2. The first law of thermodynamics is conservation of
   a) Momentum b) Energy c) Both (a) and (b) d) None of these

3. $\frac{1}{2}N_2(g)+\frac{3}{2}H_2(g)\rightarrow NH_3(g)$ $\Delta H^o =$
   a) -46.0 KJmol $^{-1}$ b) -92.3 KJmol^{-1} c) -210 KJmol^{-1} d) -601.7 KJmol^{-1}

4. Reactions that give out heat
   a) Exothermic reaction b) Endothermic reaction
   c) Free radical reaction d) All the above

5. The maximum efficiency of a steam engine operating between 100^o C and 25^{o}C is
   a) 20% b) 22.2% c) 24.8% d) 25.1%

6. Which of the following thermodynamic functions have been called " the arrow of time".
   a) Enthalpy b) Gibbs free energy c) Entropy d) Helmholtz free energy
   b)

7. The third law of thermodynamics states that is the limit $T \rightarrow 0$
   a) G=0 b) H=0 c) U=0 d) S=0

8. In the limit $T \rightarrow 0$, for a crystal
   a) $S_T=C_p/2$ b) $S_T=C_p/3$ c) $S_T=C_p/4$ d) $S_T=C_p$

9. The Gibbs phase rule is
   a) F = P – C + 2 b) F = C – P + 2 c) P = F – + 2 d) P = F – C + 1

10. The triple point is the point where
    a) Three components are in equilibrium b) The number of degrees of freedom is 3
    c) The number of degree of freedom is O d) None of the above

II. Match the following

11. Conservation of energy - Thermochemistry
12. Energy change - First law of thermodynamics
13. Efficiency of Heat engine - Componenets
14. Gibbs-Duhem equation - W/q_2
15. Phase rule - $\sum n\, id\mu\, i = 0$

SECTION – B (10X2=20)

Answer any **TEN** Questions

16. State the first law of thermodynamics.

17. Differentiate between work and energy.

18. What are phase transitions?

19. State efficiency of an engine.

20. 1g of steam is condensed at 100^oC and 1 atm. pressure Calculate the entropy change. (Given latent heat of steam is $2.238X10^6$ JKg^{-1}.).

21. State the Carnot's theorem.

22. What is the use of III law of thermodynamics?

23. Derive the Nernst heat theorem from Gibbs – Helmholtz equation.

24. What are phase boundaries?

25. Derive the variation of the Gibbs energy with temperature.

26. State the II law of Thermodynamics.

27. What is Gibbs free energy?

SECTION- C (5X8=40)

Answer any **FIVE** Questions

28. i) How is enthalpy related to the energy of the system?
    ii) The heat of formation of $C_6H_6(l)$,$Co_2(g)$ and H2O(l) at 298^oK are +11.72, - 94.05 and - 68.32k. cal per mole respectively. From these data Calculate heat of combustion of liq. Benzene. **(4+4)**

29. i) Derive Kirchhoff's equation relating the variation of enthalpy of a reaction with temperature.
    ii) What are thermo chemistry? In an endothermic reaction how do the strengths of the bonds in the reactants compare with those of their counterparts in the products? **(4+2+2)**

30. i) Heat supplied to a Carnot engine is 1897.8KJ. How much useful work can be done by the engine which works between O^0C and 100^o C?
    ii) Describe in detail the Carnot reversible cycle for establishing the maximum convertibility of heat in to work. **(3+4)**

31. .i) What is meant by chemical potential? Derive the Gibbs – Duhem equation.
    ii) Show that the entropy of any substance at low temperatures (O<T<20K), where Debye's relation for heat capacities of' crystals is valid, is one – third of the molar heat capacity. **(2+3+3)**

32. i) Determine the number of degrees of freedom is each of the following systems. Suggest the variables that could correspond to these degrees of freedom.
a) liq. water and water vapour in equilibrium.
b)Liq. Water and water vapour in equilibrium at a pressure of 1 atm.
ii) Draw and discuss the phase diagram for the water system. **(2+2+4)**

33. i) Derive Gibbs – Duhem – margules equations.
ii) State III law of thermodynamics. **(6+2)**

34. i) Establish the relationship between q_p and q_v in the synthesis of H_2O (g) from the elements assuming ideality. **(4)**
ii) Explain Helium -4 system. **(4)**

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